

Maximus Jialong Niu

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TECHNICAL SKILLS

Languages: JavaScript, TypeScript, HTML, CSS, Python, SQL, C, C++, Dart, PHP

Tools/Frameworks: React, Next.js, Redux, XState, React Query, Tailwind, Storybook, Vitest, React Testing Library, Flutter

Technologies: Git, Figma, Node.js, Sentry, AWS (S3, IAM, Lambda), Azure (DevOps, Web Apps, AD B2C), Linux

Databases: PostgreSQL (pgvector), DynamoDB

EMPLOYMENT

React Developer

Mar. 2025 – Present

Venuiti Healthcare

Waterloo ON, Canada

- Led the frontend development of the myMedcan healthcare portal using **Next.js**, **TypeScript** and **Tailwind**, integrating with a .NET core backend that supports **20,000+** registered users
- Owned the core multi-step clinical appointment booking and purchasing flow end to end, using **XState** and **React Query** for state management and data fetching, driving **15,000** bookings and **\$168,000** in purchases monthly
- Implemented custom login, sign up, and account activation flows using **Azure AD B2C** and **MSAL React**, enabling **JWT** validation via **Next.js middleware** to achieve protected routing and **100% HIPAA** and **PIPEDA** compliance
- Introduced **Storybook** to establish an isolated component development and documentation process, building a library of **200** reusable **React** components that meet **WCAG 2.2 AA** accessibility compliance throughout
- Integrated **Sentry** for real-time production error monitoring, enabling stack-trace-level visibility into client-side failures and significantly reducing mean resolution time by **35%**
- Eliminated over **180+** unnecessary component re-renders caused by auth token refreshes on the T6 web app by migrating from **React Context API** to **Redux** selectors
- Achieved **100%** unit test coverage of internal **FHIR** adapter modules on T6 using **Vitest** and **React Testing Library**

Software Developer Intern

Jan. 2023 – Apr. 2023

Wind River Systems

Ottawa ON, Canada

- Wrote and executed **50+** C/C++ unit tests on a simulated **x86** board to validate Unix utility functions ported to the **VxWorks RTOS**, ensuring **POSIX**-compliant behavior across different embedded environments
- Built an internal **Python** and **Flask** automation tool to convert between Markdown formats, processing **500+** pages and streamlining documentation workflows across teams

Full Stack Developer Intern

Jan. 2022 – Apr. 2022

My First Montessori School

Brampton ON, Canada

- Spearheaded the Montessi e-learning platform using **PHP** and **JavaScript** to enable remote learning for **150+** students

Full Stack Developer Intern

Sep. 2020 – Dec. 2020

Syntu Software

Waterloo ON, Canada

- Architected a serverless backend to handle sensitive legal evidence using **AWS DynamoDB**, **S3**, **Lambda**, and **IAM**, ensuring **100% PIPEDA** compliance

Software Developer Intern

Jan. 2020 – May. 2020

The Co-operators

Guelph ON, Canada

- Built an automation workflow utilizing **OCR** and **web scraping** to test an internal policy filing application, achieving **400+** policy submissions per hour

PROJECTS

ReWatched – Semantic search over your entire YouTube watch history

- Engineered an **MV3 Chrome extension** that semantically searches YouTube watch history by running a **RAG** pipeline on video transcripts using **OpenAI** embeddings and **IndexedDB** as a vector store

RoomEase – Agentic co-living assistant mobile app

- Built a **Flutter** app with an **agentic AI** assistant, routing natural language queries to structured in-app actions (chore delegation, message relay, schedule management) via an **OpenAI** and **Pinecone** powered **RAG** pipeline

EDUCATION

University of Waterloo

Sep. 2019 - Apr. 2024

Bachelor of Applied Science, Honours Computer Engineering

Relevant Courses: Software Architecture, Database Systems, Machine Learning, Systems Programming and Concurrency, Computer Networks, Computer Security, Compilers, Algorithms & Data Structures